

TV5 Report - Benchmark, Results, and Plots

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1. Problem

Camera-LiDAR timestamp offset creates position misalignment in ADAS fusion. This is a synthetic 2-D alignment benchmark, not a detector or tracker evaluation.

2. Method

45 conditions: scenarios A, B, C; offsets 0-200 ms; distances 10, 20, 40 m; 100 samples per condition; seed 42; noise 0.02 m per axis. Metrics are Euclidean E_{pre} , linear E_{post} , and CTRV E_{post} .

3. Benchmark

At offset 0 ms, all mean errors are 0.023-0.027 m. For A at d=20 m and 100 ms, E_{pre} =2.001 m, close to $v \times dt$ =2.000 m; linear E_{post} =0.026 m.

4. Failure case

For C at d=40 m and 100 ms, E_{pre} =1.668 m and linear E_{post} =1.351 m exceed the group threshold 0.5 m. CTRV on the same data reduces mean error to 0.025 m.

5. Engineering decision

Use CTRV when reliable offset and yaw rate are available. If offset exceeds 30 ms and yaw rate exceeds 0.2 rad/s, avoid point-to-camera-bbox association, track sensors separately, and log a warning. Prefer PTP or shared-trigger hardware synchronization.

Evidence	Location
Result table	results/results.csv
Sample data	results/samples.csv
Run log	results/run_log.txt
Plots	plots/error_vs_offset.png; plots/alignment_turn.png